

2.1 Equations of Motion

Question Paper

Course	CIE A Level Physics
Section	2. Kinematics
Topic	2.1 Equations of Motion
Difficulty	Easy

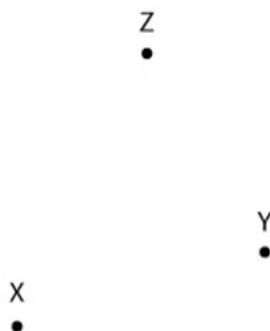
Time allowed: 20

Score: /10

Percentage: /100

Question 1

An object moves directly from X to Z



In a shorter time, a second object moves from X to Y to Z.

Which statement about the two objects is correct for the journey from X to Z?

- A. they have the same average speed
- B. they have the same average velocity
- C. they have the same displacement
- D. they travel the same distance

[1 mark]

Question 2

An experiment is done to measure the acceleration of free fall of a body from rest.

Which measurements are needed?

- A. the height of fall and the time of fall
- B. the height of fall and the weight of the body
- C. the mass of the body and the height of fall
- D. the mass of the body and the time of fall

[1 mark]

Question 3

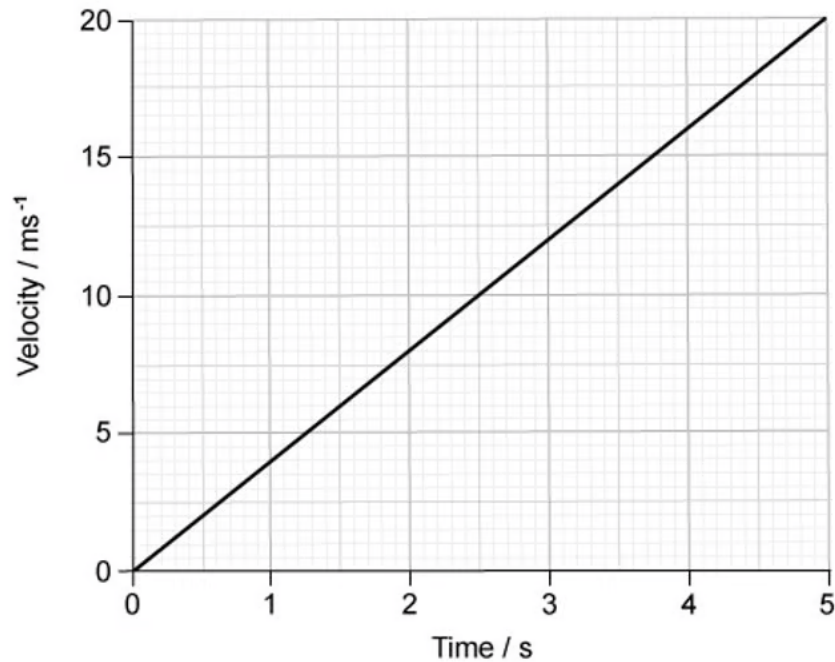
Which feature of a graph allows acceleration to be determined?

- A. the area under a displacement-time graph
- B. the area under a velocity-time graph
- C. the area under a velocity-time graph
- D. the slope of a velocity-time graph

[1 mark]

Question 4

The velocity of an object during the first five seconds of its motion is shown on the graph



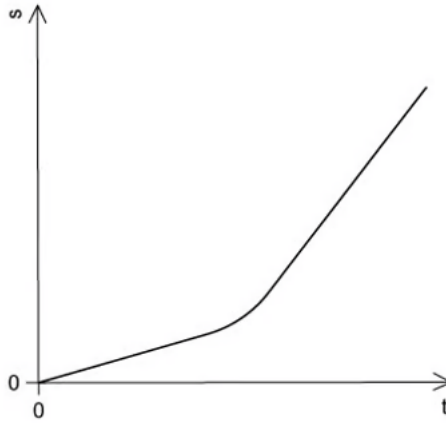
What is the displacement of the object in this time?

- A. 4 m
- B. 20 m
- C. 50 m
- D. 100 m

[1 mark]

Question 5

The variation with time t of the distance s moved by a body is shown below.



What can be deduced from the graph about the motion of the body?

- A. it accelerates continuously
- B. it starts from rest
- C. the distance is proportional to time
- D. the speed changes

[1 mark]

Question 6

An object is thrown with velocity 7.1 m s^{-1} vertically upwards on the Moon. The acceleration due to gravity on the Moon is 1.62 m s^{-2}

What is the time taken for the object to return to its starting point?

- A. 3.5 s
- B. 4.4 s
- C. 6.5 s
- D. 8.8 s

[1 mark]

Question 7

An insect jumps with an initial vertical velocity of 1.5 m s^{-1} , reaching a maximum height of $4.5 \times 10^{-2} \text{ m}$. Assume the deceleration is uniform.

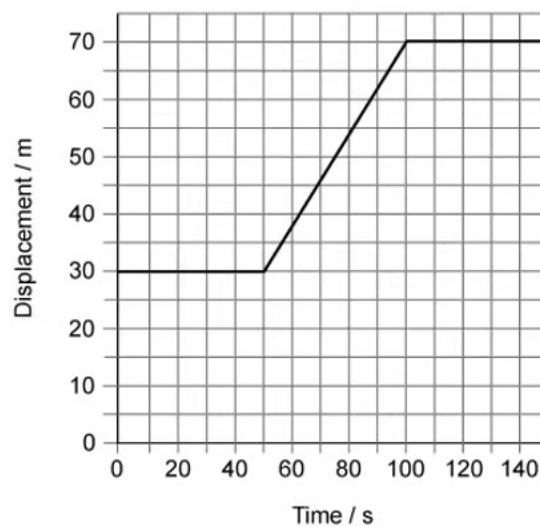
What is the magnitude of the deceleration?

- A. 6.8 m s^{-2}
- B. 9.8 m s^{-2}
- C. 25 m s^{-2}
- D. 50 m s^{-2}

[1 mark]

Question 8

A car at rest in a traffic queue moves forward in a straight line and then comes to rest again. The graph shows how the car's displacement varies with time



What is its speed while it is moving?

- A. 0.70 m s^{-1}
- B. 0.80 m s^{-1}
- C. 1.25 m s^{-1}
- D. 1.40 m s^{-1}

[1 mark]

Question 9

On a particular railway, a train driver applies the brake of the train at a yellow signal, a distance of 1.0 km from a red signal, where the train stops.

The maximum deceleration of the train is 1.1 m s^{-2}

Assuming uniform deceleration, what is the maximum safe speed of the train at the yellow signal?

- A. 2.2 m s^{-1}
- B. 20 m s^{-1}
- C. 47 m s^{-1}
- D. 220 m s^{-1}

[1 mark]

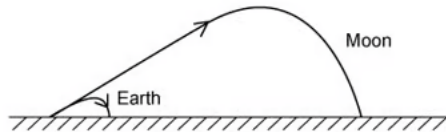
Question 10

A golf ball is hit with the same force and direction on the Earth and on the Moon.

Which diagram best represents the shapes of the paths taken by the golf ball?

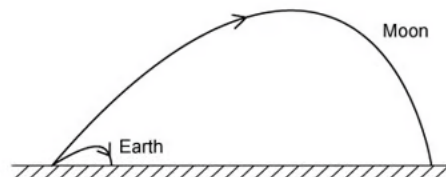
A.

A



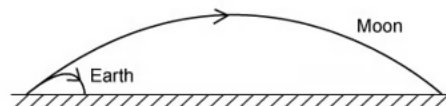
B.

B



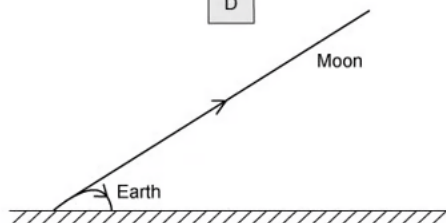
C.

C



D.

D



[1 mark]

Model Answers: Easy

1

The correct answer is **C** because:

- Displacement describes an objects change in position, measuring from its starting position to the final position
- The objects both finish at the same position so despite taking different routes, they have the same displacement

A is incorrect because the distance travelled by the second object is longer than the first one but it reaches Z in a shorter time, since average speed = total distance / total time its average speed should be larger

B is incorrect because the time taken for the second object to reach Z is less than the first and they both have the same displacement, since average velocity = displacement / time the second object has a larger velocity

D is incorrect because the first object travels directly from X to Z which is a shorter distance than X to Y to Z

2

The correct answer is **A** because:

- To determine the measurements needed to determine the acceleration of free fall, we will need to consider the equation

- $s = u t + \frac{1}{2} a t^2$

- We need to outline the quantities that we know

- $s = h$ (height)
 - $a = g$ (acceleration of free fall)
 - t (time)

- Since the falling body will initially be at rest, i.e. $u = 0$ we can simplify this to

- $h = \frac{1}{2} g t^2$

- Rearranging the equation above we can see

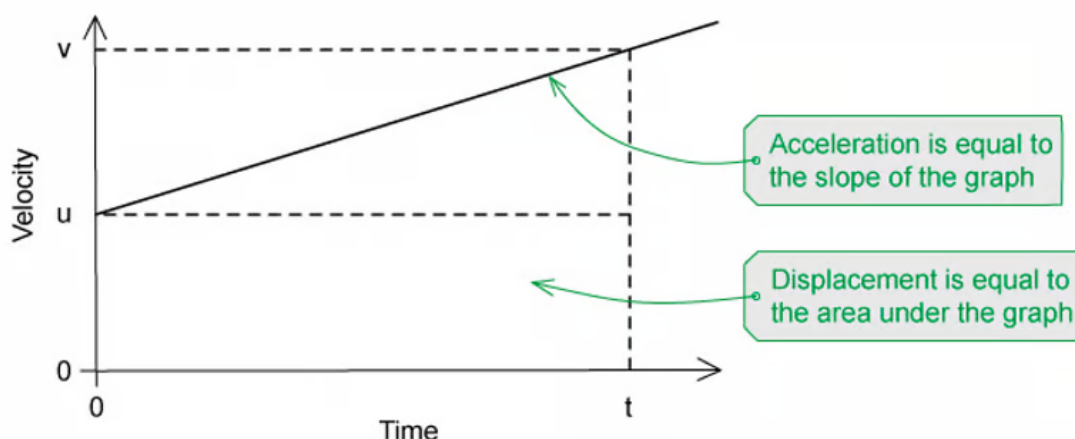
$$\circ \quad g = \frac{2h}{t^2}$$

- So, the only measurements required are the height and time of the fall

3

The correct answer is **D** because:

- Acceleration is equal to change in velocity / time
- Looking at a velocity-time graph we can see $\frac{\Delta y}{\Delta x}$ is equal to $\frac{v-u}{t}$ which is equivalent to $\frac{\text{change in velocity}}{\text{time}}$
- Therefore, acceleration is equal to the slope of the graph



A is incorrect because the area under a displacement-time graph is displacement $\times$ time which is not a physical quantity that has any use in physics

B is incorrect because the area under a velocity-time graph is equal to displacement since displacement = velocity $\times$ time

C is incorrect because the slope of a displacement-time graph is equal to velocity since velocity = displacement / time

4

The correct answer is **C** because:

- Displacement is equal to velocity $\times$ time
- This can be found from the area under the velocity - time graph
- Area under the graph = $\frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times \Delta v \times \Delta t$
- Displacement = $\frac{1}{2} \times 20 \times 5 = 50 \text{ m}$

5

The correct answer is **D** because:

- There are two clear sections on this distance-time graph
- At first the object moves with a constant speed and at a later time it accelerates briefly until reaching a higher speed which it continues to travel at for the remainder of the distance

A is incorrect because if it was accelerating continuously the line would curve upwards. Instead, it travels constantly at two different speeds

B is incorrect because it starts from a speed which is equal to the gradient of the line, if it started from rest, the line would be horizontal and curve upwards to show an acceleration

C is incorrect because the distance is proportional to time for the sections of the graph which are straight but it is not proportional the entire time

6

The correct answer is **D** because:

- First it is a good idea to write down all the quantities that we know
 - Initial velocity, $u = 7.1 \text{ m s}^{-1}$
 - $a = 1.62 \text{ m s}^{-2}$
 - the final velocity, $v = 0 \text{ m s}^{-1}$ (at the maximum height)
 - $t = ?$
- The equation that relates u , v , a and t is
 - $v = u + at$
 - $v - u = at$
 - $t = \frac{v - u}{a} = \frac{7.1}{1.62} = 4.4 \text{ s}$
- It takes 4.4 s for the object to reach its maximum height, and it will take the same

time to reach the ground since the acceleration is the same, and the final velocity will be the same, so:

- Total time = $4.4 \times 2 = 8.8 \text{ s}$

7

The correct answer is **C** because:

- First, we must write down everything we know
 - Initial velocity, $u = 1.5 \text{ m s}^{-1}$
 - Maximum height, $s = 4.5 \times 10^{-2} \text{ m}$
 - At the maximum height, the insect is at rest before it begins to move downwards, meaning the final velocity is $v = 0 \text{ m s}^{-1}$
 - $a = ?$
- The equation which relates u , v , s and a is
 - $v^2 = u^2 + 2 a s$
- Rearranging for a gives:
 - $2 a s = v^2 - u^2$
 - $a = \frac{v^2 - u^2}{2 s}$
 - $a = \frac{0 - 1.5^2}{2 \times 4.5 \times 10^{-2}} = -25 \text{ m s}^{-2}$
- The minus sign indicates the insect is slowing down, so the deceleration is equal to 25 m s^{-2}

8

The correct answer is **B** because:

- Speed is equal to $\frac{\text{displacement}}{\text{time}}$ which is given by the gradient of the graph $\frac{\Delta y}{\Delta x}$

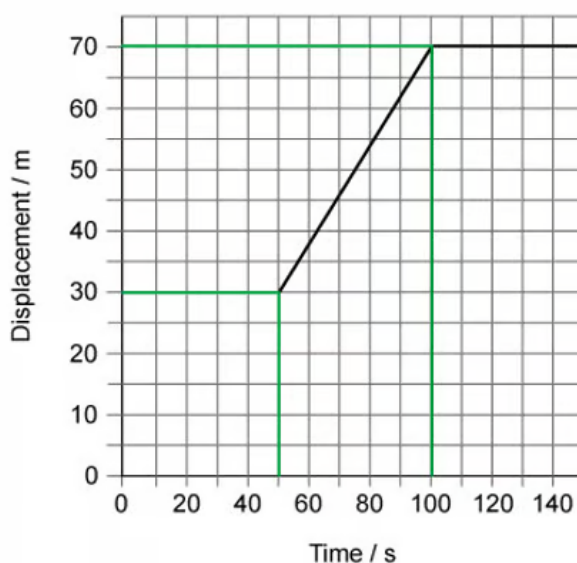
$$\circ a = \frac{0 - 1.5^4}{2 \times 4.5 \times 10^2} = -25 \text{ m s}^{-2}$$

- The minus sign indicates the insect is slowing down, so the deceleration is equal to 25 m s^{-2}

8

The correct answer is **B** because:

- Speed is equal to $\frac{\text{displacement}}{\text{time}}$ which is given by the gradient of the graph $\frac{\Delta y}{\Delta x}$
- Between 0 – 50 s and 100 – 140 s, when the lines are horizontal, the car is stationary
- Between 50 – 100 s the car is moving with constant speed and is displaced by $70 - 30 = 40 \text{ m}$
- The speed when the car is moving is equal to $\frac{40 \text{ m}}{50 \text{ s}} = 0.8 \text{ m s}^{-1}$



9

The correct answer is **C** because:

- First, we must write down the variables that we know:
 - $\circ s = 1000 \text{ m}$
 - $\circ a = -1.1 \text{ m s}^{-2}$
 - $\circ v = 0$

- $u = ?$
- The equation that links s , a , v and u is
 - $v^2 = u^2 + 2as$
 - $u^2 = v^2 - 2as = 0 - (2 \times (-1.1) \times 1000) = 2200$
 - $u = \sqrt{2200} = 46.9 \text{ m s}^{-1}$

10

The correct answer is **C** because:

- When the golf ball is struck upwards, the force of gravity will always bring it back to the surface
- The force of gravity is much higher on the Earth so the golf ball is pulled towards the surface faster which is why it has a much shorter range than the ball on the Moon
- The shape of the golf ball's path depends on air resistance – on Earth, gravity pulls the ball down while air resistance opposes its motion leading to a path which is not symmetrical
- On the Moon, however, there is no atmosphere, so there is no air resistance acting on the ball. This means the path of the golf ball will be symmetrical, as shown in C

A & B are incorrect because these paths are asymmetrical which implies there is an opposing force acting on the ball, such as air resistance, but since there is no atmosphere on the Moon, this is not possible

D is incorrect because the gravity on the Moon is small but it is not zero, so the ball would return to the surface

